A QUICK REFERENCE GUIDE TO USING Python FUNCTIONS

Summary

The Statistics Team members within the Surveillance Branch (SB), DHQP, have developed a series of statistical comparison functions. These functions were developed to conduct statistical tests for epidemiological measures, consistent with the methods used by NHSN. Measures tested include common epidemiological measures such as incidence density rates, proportions, and ratios such as standardized metrics known as the standardized infection ratio (SIR), standardized utilization ratio (SUR), standardized antimicrobial administration ratio (SAAR), etc.

Overview

Six Python functions created by NHSN statisticians are listed below:

Description	Program Filename	Function Name
Testing SIR=1	MID-P FOR SIR EQUAL TO 1	SIRcomp
Comparing 2 SIRs (ratio)	MID-P FOR COMPARING 2 SIRs	binom
95%CI of incidence density rate (single rate)	MID-P FOR INCIDENCE DENSITY RATE	rateCIComp
Comparing 2 rates (ratio)	MID-P FOR COMPARING 2 RATES	two_rates
95%CI of single proportion	MID-P FOR SINGLE PROPORTION	binom_ci
Comparing 2 proportions (ratio)	MID-P FOR COMPARING 2 PROPORTIONS.	TWOBYTWO

To facilitate user application, a brief description, accompanying example data and corresponding outputs are provided below for each function.

1. MID-P FOR SIR EQUAL TO 1

Description

This function can be used to test whether SIR (or similar standardized metrics like SUR, SAAR) is statistically different from 1, consistent with the methods used by NHSN.

Example dataset and Input parameters

Input parameters:

OBS=observed value, EXP=predicted value

```
# Create a sample dataframe
df = pd.DataFrame(
    {'obs': [1,0, np.nan,1],
    'pred': [2.3, 3, 5, 0.6]}
    )

# Apply SIRcomp function to each row
df[['SIR_pval','SIR', 'SIR_L', 'SIR_U']] = df.apply(
    lambda row: pd.Series(SIRcomp(row['obs'], row['pred'])),
    axis=1)
```

Outputs and interpretations

Abbreviations

obs = observed value

exp = predicted value

SIR_pval = 2 tailed mid-P value

SIR_L = lower limit of 95% CI

SIR_U = upper limit of 95% CI

Note:

*No results will be generated if entry values are missing or impossible values.

*When SIR=0, the lower limit (SIR_L) of 95%CI is set to missing by default. A user may overwrite SIR_L=0 if desired.

	obs	pred	SIR_pval	SIR	SIR_L	SIR_U
0	1.0	2.3	0.4311	0.435	0.022	2.144
1	0.0	3.0	0.0498	0.000	NaN	0.999
2	NaN	5.0	NaN	NaN	NaN	NaN
3	1.0	0.6	NaN	NaN	NaN	NaN

First example (First row of output):
The p-value is 0.4311. Because this p-value is greater than 0.05, we would conclude that computed SIR value=0.435 is not statistically different from 1.0. 95%CI also includes the null (1.0)

Second example:
Observed value is 0 so that SIR=0 in which the lower limit is set to missing by default.

Third example:
Observed value is missing so that SIR and 95%CI are not computed.

Fourth example:
Predicted value is <1, so that SIR and 95%CI are not computed.

REFERENCE:

1. JH ABRAMSON. WINPEPI PROGRAMS. DESCRIBE MANUAL (VERSION 2.42), PAGE-52. Available at 'http://www.brixtonhealth.com/pepi4windows.html'
2. Dean AG, Sullivan KM, Soe MM. OpenEpi: Open Source Epidemiologic Statistics for Public Health, Version. www.OpenEpi.com.
3. Rothman KJ, Boice JD Jr: Epidemiologic analysis with a programmable calculator. NIH Pub No. 79-1649. Bethesda, MD: National Institutes of Health, 1979;31-32.
4. Rothman KJ, Greenland S. Modern Epidemiology, 2nd Edition. Lippincott-Raven Publishers, Philadelphia, 1998.
5. Geoffrey rc, shu-ying y. Mid-p confidence intervals for the poisson expectation. Statistics in medicine, 13,2189-2203 (1994)

2. MID-P FOR COMPARING 2 SIRs

Description
This function can be used to perform a comparison between two SIRs, consistent with the methods used by NHSN. SIR could be substituted by similar standardized metrics like SUR, SAAR.

Example dataset and input parameters
Data input layout:
o1= observed events in group1
e1= predicted events in group1
o2= observed events in group2
e2= predicted events in group2

```
# Sample DataFrame
data = pd.DataFrame({
    "o1": [1,1,1,3],  # Observed events in group 1
    "e1": [1,0.5,2,np.nan],  # predicted events in group1
    "o2": [2,1,0,4],  # Observed events in group 2
    "e2": [3,2,5.2, 5.2]  # predicted events in group2
})

# Apply the function to each row
data[['midP', 'RATIO', 'LL', 'UL']] = data.apply(
    lambda row:pd.Series(binom(row["o1"], row["e1"], row["o2"], row["e2"])),
    axis=1)

print(data)
```

Outputs and interpretations

Abbreviations
midP= 2 tailed mid-P value
RATIO= SIR in group2 / SIR in group1 = SIR2/SIR1 = (&O2/&E2)/(&O1/&E1)
LL= 95% lower bound of RATIO
UL= 95% Upper bound of RATIO
Note: When RATIO=0, the lower limit (LL) of 95%CI is set to missing by default. A user may change it to LL=0 if desired.

o1	e1	o2	e2	midP	RATIO	LL	UL
1	1.0	2	3.0	0.734375	0.666667	0.050708	19.662395
1	0.5	1	2.0	NaN	NaN	NaN	NaN
1	2.0	0	5.2	0.277778	0.000000	NaN	7.307810
3	NaN	4	5.2	NaN	NaN	NaN	NaN

First comparison (first row): SIR for group1 is 1/1 and SIR for group2 is 2/3. The ratio of 2 SIRs (group2 SIR/ group1 SIR) is shown as 0.667 with 95%CI. As the midP value is >0.05, we would conclude that there is NO significant difference between the two SIRs.

Second example shows predicted number<1 for group1 so that mid-p and ratio are not computed.

Third example has ratio of 0 so that the lower limit of 95%CI is not computed.

Fourth example has missing predicted value for group1 so that mid-p and ratio are not computed.

Reference

1. Breslow NE, Day NE. Statistical methods in cancer research. Volume II--The design and analysis of cohort studies. IARC Sci Publ. 1987;(82):1-406. PMID: 3329634. http://www.iarc.fr/en/publications/pdfs-online/stat/sp82/SP82_vol2-3.pdf
2. Erich L. Lehmann, Joseph P. Romano. Testing Statistical Hypotheses. Wiley, New York, 1959 (Third edition, 2005, Springer Texts in Statistics)
3. John Pezzullo. Interactive Statistical Calculation Pages. www.Statpages.com.
4. Dean AG, Sullivan KM, Soe MM. OpenEpi: Open Source Epidemiologic Statistics for Public Health, Version. www.OpenEpi.com, updated 2013/03/21

3. MID-P FOR INCIDENCE DENSITY RATE

Description

This function can be used to perform confidence interval testing around a single incidence density rate (e.g., CLABSI rate), consistent with methods used by NHSN.

Example dataset and input parameters

Input parameters:

numer=numerator, denom=denominator.

```
# Sample DataFrame
data = pd.DataFrame({
    "numer": [0, 10],
    "denom": [10, 10]
})
```

```
# Apply the function to each row
data[['rate', 'rate_l', 'rate_u']] = data.apply(
    lambda row:pd.Series(rateCIComp(row["numer"], row["denom"])),
    axis=1)
print(data)
```

Outputs and interpretations

The rate here is multiplied by 1000 so that Incidence density rate is per 1000 person-time & rate_l & rate_u represents.lower and upper limit of 95%CI respectively.
Note: When rate=0, the lower limit (rate_l) of 95%CI is set to missing by default. A user may change it to rate_l=0 if desired.

numer	denom	rate	rate_l	rate_u
0	10	0.0	NaN	299.573210
10	10	1000.0	507.940491	1782.488654

First example (First row of output):
As the incidence density rate=0, the lower limit (rate_l) of 95%CI is set to missing by default.

Second row:
Incidence density rate is 1,000 per 1,000 person-time along with 95%CI.

Reference

1. Dean AG, Sullivan KM, Soe MM. OpenEpi: Open Source Epidemiologic Statistics for Public Health, Version 2.3.1. www.OpenEpi.com
2. Rothman KJ, Boice JD Jr: Epidemiologic analysis with a programmable calculator. NIH Pub No. 79-1649. Bethesda, MD: National Institutes of Health, 1979;31-32.
3. Rothman KJ, Greenland S. Modern Epidemiology, 2nd Edition. Lippincott-Raven Publishers, Philadelphia, 1998.

4. MID-P FOR COMPARING 2 RATES

Description

This function can be used to perform a comparison between two incidence density rates (e.g., CAUTI rates), consistent with the methods used by NHSN.

Example dataset and input parameters
Input parameters
O1= observed events in group-1
PT1=person-time in group-1
O2= observed events in group-2
PT2= person-time in group-2

```
# Sample DataFrame
data = pd.DataFrame({
    "O1": [np.nan, 1,1,1],  # Observed events in group 1
    "PT1": [2.3, 2,2,2],  # Person-time for group 1
    "O2": [2,3,3,0],  # Observed events in group 2
    "PT2": [4.5, -4, 4,5]  # Person-time for group 2
})

# Apply the function to each row
data[['MID_P', 'RATE_RATIO', 'LL', 'UL']] = data.apply(
    lambda row:pd.Series(two_rates(row["O1"], row["PT1"], row["O2"], row["PT2"])),
    axis=1)

print(data)
```

Outputs and interpretations
Abbreviations
mid-P=mid p-value for comparing between 2 rates
RATE_RATIO = Rate in group-2 / Rate in group-1
LL=lower limit of 95%CI of rate ratio
UL=Upper limit of 95%CI of rate ratio
Note: When RATE_RATIO=0, the lower limit (LL) of 95%CI is set to missing by default. A user may change it to LL=0 if desired.

O1	PT1	O2	PT2	MID_P	RATE_RATIO	LL	UL
NaN	2.3	2	4.5	NaN	NaN	NaN	NaN
1.0	2.0	3	-4.0	NaN	NaN	NaN	NaN
1.0	2.0	3	4.0	0.790123	1.5	0.159879	39.497559
1.0	2.0	0	5.0	0.285714	0.0	NaN	7.600122

First and second examples show missing value in group1 and negative value in group2 respectively so that the function would not generate any statistical output.

The third and fourth example show midP >0.05, we would conclude that there is no significant difference between 2 incidence rates.

Reference

1. Erich L. Lehmann, Joseph P. Romano. Testing Statistical Hypotheses. Wiley, New York, 1959(Third edition, 2005, Springer Texts in Statistics)
2. John Pezzullo. Interactive Statistical Calculation Pages. www.Statpages.com.
3. Dean AG, Sullivan KM, Soe MM. OpenEpi: Open Source Epidemiologic Statistics for Public Health, Version. www.OpenEpi.com, updated 2013/03/21
4. Austin, Harland (Emory). Epidemiology method-II: Statistical issues for Density type follow-up studies.

5. MID-P FOR SINGLE PROPORTION

Description

This function can be used to perform confidence interval testing around a single proportion (e.g., healthcare worker flu vaccination percentages). This macro can also be used to compare a single proportion to a benchmark value, consistent with methods used by NHSN

Example dataset and input parameters

Input parameters:

vx=numerator, vn=denominator.

```
# Sample DataFrame
data = pd.DataFrame({
    "num": [1, 10, np.nan],  # Numerator
    "denom": [2,4,0.03]  # Denominator
})


# Apply the function to each row
data[['vp', 'dl', 'du']] = data.apply(
    lambda row:pd.Series(binom_ci(row["num"], row["denom"])),
```

```
    axis=1)


print(data)
```

Outputs and interpretations

Abbreviations
vp=proportion,
dl=lower limit of 95%CI
du=upper limit of 95%CI

Note: When vp=0, the lower limit (dl) of 95%CI is set to missing by default. A user may change it to dl=0 if desired.

num	denom	vp	dl	du
1.0	2.00	0.5	0.024998	0.975002
10.0	4.00	NaN	NaN	NaN
NaN	0.03	NaN	NaN	NaN

First example: Point estimate of single proportion of 1/2 is 0.5 with 95%CI provided.

Second example has denominator smaller than numerator so that there would be no statistical output.

Third example has missing numerator so that there would be no statistical output.

Reference

1. Dean AG, Sullivan KM, Soe MM. OpenEpi: Open Source Epidemiologic Statistics for Public Health, Version 2.3.1. www.OpenEpi.com
2. Rothman KJ, Boice JD Jr: Epidemiologic analysis with a programmable calculator. NIH Pub No. 79-1649. Bethesda, MD: National Institutes of Health, 1979;31-32.
3. Fleiss JL. Statistical Methods for Rates and Proportions, 2nd Ed. John Wiley & Sons, New York, 1981.
4. Newcombe RG. Two-sided confidence intervals for the single proportion: comparison of seven methods. Statistics in Medicine 1988;17:857-872.
5. Stein Vollset. Confidence intervals for a binomial proportion statistics in medicine,12,809-824 (1993)

6. MID-P FOR COMPARING 2 PROPORTIONS

Description

This function can be used to perform a comparison of two proportions (2 x 2 contingency table), ie, to compare the risks (prevalence) of outcome in two independent groups.

Example dataset and input parameters

Input parameters:

- exposed with disease (A),
- not-exposed with disease (B),
- exposed without disease (C),
- not-exposed without disease (D).

Letters refer to positions in classic 2X2 exposure/disease status table.

```
Data input layout:
A= E+,D+
B= E-,D+
C= E+,D-
D= E-,D-
                                  Disease/Outcome
                          |    Yes                 No            |
                -----------|-----------------------------------|-------
        Exposure     Yes  |     A                   C            |
                     No   |     B                   D            |
                -----------|-----------------------------------|-------
```

```
# sample dataframe
data=pd.DataFrame({
   "A": [10,20,3,-3],
   "B": [20,60,5,5],
   "C": [11,80,7, np.nan],
   "D": [21,100,np.nan,7]
})

# Apply the function to each row
data['midP']=data.apply(
   lambda row:pd.Series(TWOBYTWO(row["A"], row["B"], row["C"], row["D"])),
   axis=1)

print(data)
```

Outputs and interpretations

A	B	C	D	midP
10	20	11.0	21.0	0.934208
20	60	80.0	100.0	0.002752
3	5	7.0	NaN	NaN
-3	5	NaN	7.0	NaN

First example shows midP=0.934 indicates that proportion (prevalence, risk) of disease in exposed group is NOT different from that of non-exposed group.

The second example shows mid-p=0.00275 indicating that proportion (prevalence, risk) of disease in exposed group (A/(A+C)) is statistically different from that of non-exposed group (B/(B+D)).

Other examples have missing cell value or incorrect value such as negative value, mid-P is not computed.

Reference

1. Dean AG, Sullivan KM, Soe MM. OpenEpi: Open Source Epidemiologic Statistics for Public Health, Version 2.3.1. www.OpenEpi.com
2. Rothman KJ, Boice JD Jr: Epidemiologic analysis with a programmable calculator. NIH Pub No. 79-1649. Bethesda, MD: National Institutes of Health, 1979;31-32.
3. Fleiss JL. Statistical Methods for Rates and Proportions, 2nd Ed. John Wiley & Sons, New York, 1981